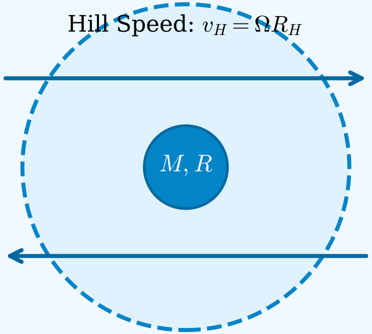
Goldreich, Lithwick, & Sari (2004) Planetesimal Coagulation Physics Architecture

1. Hill Sphere & Shear Flow

Hill Radius: $R_H = a(M/3M_*)^{1/3}$

Hill Speed: $v_H = \Omega R_H$



Keplerian Shear: $\Delta v_K \sim \Omega x$

Geometric Ratio: $\alpha \equiv R/R_H \ll 1$

Velocity Dispersion: $\theta \equiv u/v_H$

Scale Height: $H \sim u/\Omega \approx \theta R_H$

Escape Speed: $v_{\rm esc} \approx \sqrt{6}\,\alpha^{-1/2} v_H$

2. Coagulation Regimes & Growth

2D Shear-Dominated ($\theta \le \alpha^{1/2}$)

$\mathrm{d}R/\mathrm{d}t \sim \frac{\Sigma\Omega}{\rho}\alpha^{-3/2}$ (Ultra-cold thin disk)

3D Shear-Dominated ($\alpha^{1/2} < \theta \le 1$)

$\mathrm{d}R/\mathrm{d}t \sim \frac{\Sigma\Omega}{\rho}\alpha^{-1}\theta^{-1}$ (3-body shear boost)

Dispersion Focused ($1 < \theta \le \alpha^{-1/2}$)

$\mathrm{d}R/\mathrm{d}t \sim \frac{\Sigma\Omega}{\rho}\alpha^{-1}\theta^{-2}$ (2-body Safronov)

\textbf{Geometric Unfocused} ($\theta > \alpha^{-1/2}$)

$\mathrm{d}R/\mathrm{d}t \sim \frac{\Sigma\Omega}{\rho}$ (Direct physical collisions)

3. Velocity Equilibrium

\textbf{Viscous Stirring}:

$\left(\frac{\mathrm{d}u^2}{\mathrm{d}t}\right)_{\rm stir} \sim \frac{\Sigma_{\rm olig}}{\Sigma}\Omega v_H^2$

\textbf{Collisional Damping}:

$\left(\frac{\mathrm{d}u^2}{\mathrm{d}t}\right)_{\rm damp} \sim -\frac{\Sigma\Omega}{\rho r}u^2$

\textbf{Equilibrium} $\theta_{\rm eq}$:

$\theta_{\rm eq} \sim \left(\frac{\Sigma_{\rm olig} r}{\Sigma R \alpha}\right)^{1/4} \sim 0.1 - 0.3$

\textbf{Outer Planet Timescales}

- Uranus (19.2 AU): **9.8** Myr

  (vs. Safronov: 587 Myr)

- Neptune (30.1 AU): **23.3** Myr

  (vs. Safronov: 1.4 Gyr)